

Arabic Dialect Identification Using Logistic Regression (TD-IDF) and MARBERT

Applied Natural Language Processing Project
Dr.Maram Almaghrabi

Student Name:	Student ID:	Signature:
Reuf bakr alshrif	443002428	
Jana hesham alhlees	443002347	
Mayar turki alowaydhi	443003550	

Table of Contents

1. Introduction 3
2. Objectives 4
3. Data Overview 4
4. Data Description 5
5. Data Preprocessing 6
5.1 Understanding the Dataset.....6
5.2 Data Cleaning and Text Normalization 6
5.3 Tokenization.....7
5.4 Label Encoding of Dialect Classes.....7
5.5 Train–Validation Split Strategy 7
6. Methodology.....8
6.1 Machine Learning Model: Logistic Regression TF-IDF 8
6.2 Deep Learning Model: MARBERT Fine-Tuning 8
6.3 Evaluation Procedure.....10
7. Results 10
7.1 Logistic Regression TF-IDF Results.....10
7.2 MARBERT Results 11
7.3 Interactive Web Interface (the MARBERT model) 13
7.4 Comparative Interpretation 14
7.5 Summary of Findings 15
8. conclusion 15
9. References.....16
10. Appendix A: Google Colab Implementation 16

1. Introduction

Arabic is one of the most widely spoken languages in the world and is characterized by a rich diversity of regional dialects that differ significantly in vocabulary, pronunciation, and grammatical structure. These dialects are widely used in daily communication, social media, and informal written texts, which makes automatic Arabic dialect identification an important and challenging task in the field of Natural Language Processing (NLP).

Automatic Arabic dialect detection plays a crucial role in many real-world applications, including sentiment analysis, social media monitoring, chatbots, recommendation systems, and text analytics. However, this task remains challenging due to several factors such as the absence of standardized spelling for dialects, strong similarities between some regional varieties, and the imbalance in labeled data across dialect classes.

This project aims to develop an intelligent system for Arabic dialect detection using both traditional machine learning and deep learning approaches. Two models are implemented and compared: a TF-IDF–based Logistic Regression classifier as a baseline model, and MARBERT, a transformer-based deep learning model specifically designed for Arabic dialectal text. The performance of both models is evaluated using standard classification metrics such as Accuracy and Macro F1-score. In addition, an interactive chatbot-style interface is developed to allow users to input Arabic sentences and receive real-time dialect predictions, demonstrating the practical applicability of the proposed system.

2. Objectives

The main objective of this project is to develop an intelligent system for automatic Arabic dialect detection from written text using machine learning and deep learning techniques. This is achieved through the following specific objectives:

1. To build an automatic classification system capable of identifying Arabic dialects from text data with good accuracy.
2. To implement a traditional machine learning model using Logistic Regression with TF-IDF as a baseline classifier.
3. To implement a deep learning model based on MARBERT to improve dialect classification performance.
4. To apply appropriate data cleaning and text preprocessing techniques to enhance data quality.
5. To evaluate the performance of the models using standard metrics such as Accuracy and Macro F1-score.
6. To conduct a comparative analysis between the traditional and deep learning models to determine the most effective approach.
7. To develop an interactive chatbot interface that allows users to input Arabic text and receive real-time dialect predictions

3. Data Overview

This project is based on a dataset consisting of Arabic text written in various regional dialects, collected from multiple real-world sources. The dataset reflects natural daily language usage and is suitable for the task of automatic Arabic dialect identification due to its linguistic diversity.
The data was used to train and evaluate classification models for distinguishing between different Arabic dialects. The textual content was selected as the main input feature, while the corresponding dialect was used as the target class. The diversity of dialects within the dataset supports the development of a robust model capable of generalizing across different regional language variations.

4. Data Description

The dataset used in this project consists of **112,000 records with 7 features** *(Dataset shape: 112,000 × 7)* representing real-world Arabic dialectal text. Each record corresponds to a short Arabic sentence reflecting daily conversational usage across different Arab countries

The original dataset includes the following columns: lang, sent, city, country_id, country, city_id, and source. Since the main objective of this project is to perform automatic dialect classification based solely on textual content, only two columns were selected for modeling:

- sent, which represents the input text, and
- country, which represents the target dialect label.

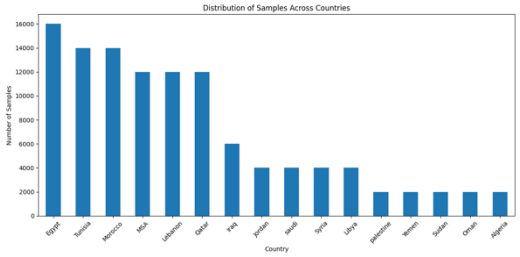
The dataset covers 16 different dialect classes, including: Saudi, Egyptian, Moroccan, Tunisian, Iraqi, Syrian, Lebanese, Jordanian, Palestinian, Qatari, Omani, Yemeni, Sudanese, Libyan, Algerian, in addition to Modern Standard Arabic (MSA). This makes the task a multi-class classification problem.

Since this task is formulated as a multi-class classification problem, the dataset was divided into training and validation/testing subsets to ensure reliable evaluation of the system's generalization performance. The split was applied during the model development and performance assessment stages.
The distribution of samples across different countries reveals a clear class imbalance, where countries such as Egypt and Morocco contain a larger number of samples, while others such as Palestine, Yemen, and Sudan are represented with fewer samples. This imbalance was taken into consideration during the model training phase.

```
Dataset shape: (112000, 7)
Columns: ['lang', 'sent', 'city', 'country_id', 'country', 'city_id', 'source']

First 5 rows:
```

	lang	sent	city	country_id	country	city_id	source
0	JER	هناك، قدام مكتب استعلامات السياح بالزيط.	Jerusalem	11	palestine	0	MADAR
1	JER	عمري ما سمعت هدا العنوان هون.	Jerusalem	11	palestine	0	MADAR
2	JER	خليك ماشي دغري لتلاقي صيدلية.	Jerusalem	11	palestine	0	MADAR
3	JER	قديش الفطور؟	Jerusalem	11	palestine	0	MADAR
4	JER	كيف بقدر أساعدك؟	Jerusalem	11	palestine	0	MADAR



5. Data Preprocessing

5.1 Understanding the Dataset

We started by exploring the dataset, which contains short and medium-length Arabic text samples labeled by dialect. The dataset includes a wide variety of Arabic dialects such as Saudi, Egyptian, Gulf, Maghrebi, and Sudanese dialects. This diversity makes the classification task both realistic and challenging.
Each data record originally consisted of multiple columns, including:
lang, sent, city, country_id, country, city_id, and source.
For the purpose of dialect classification, only the text column (sent) and the target label (country) were selected for modeling.

5.2 Data Cleaning and Text Normalization

Before training the models, several preprocessing steps were applied to clean the dataset and improve data quality:

- Rows with **missing text or missing labels** were removed to ensure consistency.
- All text values were converted to **string format** to avoid data type issues.
- A **light text normalization process** was applied, including:
  - Removing Arabic diacritics.
  - Normalizing different variants of letters (such as [إ،أ،آ،ا], 'ا'
  - and 'ى', 'ي').
  - Removing non-Arabic symbols and extra spaces.

Importantly, **dialectal spelling variations were intentionally preserved**, as they carry important linguistic cues that help distinguish regional dialects.

```
import re


df = df.dropna(subset=[TEXT_COL, LABEL_COL]).reset_index(drop=True)


df[TEXT_COL] = df[TEXT_COL].astype(str)


def clean_text(text):
    text = re.sub(r'[\u0617-\u061A\u064B-\u0652]', '', text)
    text = re.sub(r'[إأآا]', 'ا', text)
    text = re.sub(r'ى', 'ي', text)
    text = re.sub(r'ة', 'ه', text)
    text = re.sub(r'[^\u0600-\u06FF\s]', '', text)
    text = re.sub(r'(.)\1+', r'\1', text)
    text = re.sub(r'\s+', ' ', text).strip()
    return text

df["clean_sent"] = df[TEXT_COL].apply(clean_text)
```

5.3 Tokenization

As part of the preprocessing stage, tokenization was applied automatically during feature extraction using the TF-IDF vectorizer and during text preparation for the MARBERT model. This step splits the input text into meaningful word-level and sub-word units, allowing both the traditional machine learning model and the transformer-based model to effectively learn linguistic patterns from the data.

5.4 Label Encoding of Dialect Classes

Since machine learning models cannot directly process categorical text labels, **Label Encoding** was applied to convert country (dialect) names into numeric form.
Each of the **16 dialect classes** was assigned a unique integer using the *Scikit-learn* LabelEncoder.

For example:
(Algeria → 0, Egypt → 1, Iraq → 2, … , Saudi → 15).

This transformation enables the model to perform mathematical operations on labels while preserving the original mapping between numeric values and dialect classes for interpretation.

```
# Label encoding

le = LabelEncoder()
df['label_encoded'] = le.fit_transform(df[LABEL_COL])
label2id = {label: idx for idx, label in enumerate(le.classes_)}
id2label = {v: k for k, v in label2id.items()}

print("\nLabel mapping (label -> id):")
print(label2id)
```

```
Label mapping (label -> id):
{'Algeria': 0, 'Egypt': 1, 'Iraq': 2, 'Jordan': 3, 'Lebanon': 4, 'Libya': 5, 'MSA': 6, 'Morocco': 7, 'Oman': 8, 'Qatar': 9, 'Sudan': 10, 'Syria': 11, 'Tunisia': 12,
```

5.5 Train–Validation Split Strategy

To ensure fair evaluation and balanced representation of all dialect classes during training and testing, a **stratified data split** was applied as follows:

- **85%** of the data was used for **training**
- **15%** of the data was used for **validation**

This strategy preserved the original distribution of dialect classes and ensured that both sets contained samples from all dialect categories.

```
Train shape: (95200, 2)
Val shape: (16800, 2)
```

The preprocessing steps were carefully selected to improve data quality while preserving important dialectal characteristics. Data cleaning and normalization reduce noise and inconsistencies in the text, while label encoding enables effective model training. Tokenization ensures that the textual input is converted into meaningful numerical representations suitable for both machine learning and deep learning models. At the same time, dialect-specific spelling variations were intentionally preserved because they carry essential linguistic information for accurate dialect discrimination.

6. Methodology

This project followed a structured pipeline designed to compare a traditional Machine Learning model with a modern Deep Learning model for Arabic dialect identification. The process consisted of several clear steps, beginning with understanding the dataset, preparing the text, and building two classification models: TF-IDF with Logistic Regression, and MARBERT fine-tuning.

6.1 Machine Learning Model: Logistic Regression TF-IDF

As a baseline, we built a traditional Machine Learning model.

TF-IDF Feature Extraction

The text was converted into numeric features using TF-IDF, where the vectorizer considered both unigrams and bigrams. This helped capture short expressions and dialect-specific word patterns such as "شنية"، "وين"، "ايه يا عم".

We limited the vocabulary to 50,000 features to maintain efficiency while keeping important dialectal variations.

Logistic Regression Classification

Logistic Regression was used as the classifier due to its strong performance on text classification tasks. We enabled class balancing and increased the number of iterations to ensure stable convergence.

This model served as a fast and interpretable baseline for comparison.

6.2 Deep Learning Model: MARBERT Fine-Tuning

The second model in our comparison was MARBERT, a transformer model trained specifically on Arabic dialects.

Text samples were tokenized using the MARBERT tokenizer with a maximum length of 64 tokens. This length aligns well with the typical size of dialectal sentences in the dataset.
Training Configuration:

MARBERT was fine-tuned with the following settings:

- Learning rate: 1e-5
- Batch size: 16
- Gradient accumulation: 2 steps
- Mixed-precision (FP16) training for speed
- Two epochs, as the model converges quickly on text classification tasks

This setup allowed MARBERT to learn contextual, semantic, and morphological features crucial for dialect discrimination.

```
# Tokenizer & Model (MARBERT)

MODEL_NAME = "UBC-NLP/MARBERT"

tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME, use_fast=True)

# Tokenize function
def preprocess(batch):
    return tokenizer(batch["text"], truncation=True, max_length=128)

tokenized_datasets = datasets.map(preprocess, batched=True)

# Data collator (handles dynamic padding)
data_collator = DataCollatorWithPadding(tokenizer=tokenizer)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
tokenizer_config.json: 100% 376/376 [00:00<00:00, 34.0kB/s]
config.json: 100% 701/701 [00:00<00:00, 75.6kB/s]
vocab.txt: 1.10M/? [00:00<00:00, 20.6MB/s]
special_tokens_map.json: 100% 112/112 [00:00<00:00, 13.8kB/s]
Map: 100% 95200/95200 [00:07<00:00, 10304.52 examples/s]
Map: 100% 16800/16800 [00:00<00:00, 18639.83 examples/s]
```

```
# TrainingArguments & Trainer

output_dir = "./marbert_dialect_model"

os.environ["WANDB_DISABLED"] = "true"

training_args = TrainingArguments(
    output_dir=output_dir,
    eval_strategy="epoch",
    save_strategy="epoch",
    learning_rate=1e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=32,
    weight_decay=0.02,
    num_train_epochs=2,
    load_best_model_at_end=True,
    metric_for_best_model="f1_macro",
    save_total_limit=2,
    fp16=torch.cuda.is_available(),  # use mixed precision if CUDA available
    logging_dir="./logs",
    logging_steps=200,
    optim="adamw_torch" # Explicitly use PyTorch's AdamW implementation to avoid 'fused=True' issue
)

# Early stopping callback
callbacks = [EarlyStoppingCallback(early_stopping_patience=1)]

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized_datasets["train"],
    eval_dataset=tokenized_datasets["validation"],
    tokenizer=tokenizer,
    data_collator=data_collator,
    compute_metrics=compute_metrics,
    callbacks=callbacks
)
```

6.3 Evaluation Procedure

Both models were evaluated using:

- **Accuracy**
- **Macro F1 Score**
- **Confusion Matrix**

The Macro F1 metric was especially important because the dataset contains dialect classes of varying sizes.Confusion matrices were later used to analyze misclassification patterns and understand which dialects were most challenging.

7. Results

After both models were trained, we compared their performance using the validation set.

7.1 Logistic Regression TF-IDF Results

The TF-IDF + Logistic Regression model showed reasonable performance:

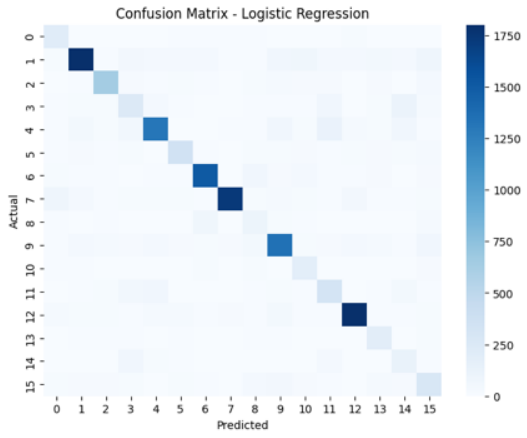
- **Accuracy:** 72%
- **Macro F1:** 60%

While the model was able to capture some frequent dialectal words and expressions, it struggled in scenarios where dialects share similar vocabulary or where the sentence was too short to contain clear markers.

The confusion matrix revealed:

- Frequent overlap between Gulf dialects
- High confusion within North African dialects (Morocco, Tunisia, Algeria)
- Difficulty distinguishing neutral or Modern Standard Arabic content

This confirmed that traditional lexical features alone are not sufficient for fine-grained dialect identification.



To further evaluate the baseline Logistic Regression model, several unseen Arabic text samples were tested. The model was able to generate dialect predictions for all inputs, demonstrating its ability to generalize beyond the training data. However, some misclassifications were observed, particularly between closely related Gulf dialects. This behavior reflects the limitations of traditional machine learning approaches when dealing with subtle linguistic variations. Despite that, the model still provides a reasonable baseline for comparison with deep learning models.

```
Logistic Regression Results
Accuracy: 0.7243
F1 Macro: 0.6091

Classification Report:
```

	precision	recall	f1-score	support
Algeria	0.47	0.68	0.56	300
Egypt	0.87	0.75	0.81	2400
Iraq	0.75	0.72	0.73	900
Jordan	0.40	0.41	0.41	600
Lebanon	0.81	0.73	0.77	1800
Libya	0.61	0.58	0.60	600
MSA	0.87	0.84	0.85	1800
Morocco	0.94	0.83	0.88	2100
Oman	0.26	0.35	0.30	300
Qatar	0.77	0.75	0.76	1800
Sudan	0.37	0.60	0.46	300
Syria	0.45	0.53	0.49	600
Tunisia	0.88	0.86	0.87	2100
Yemen	0.40	0.64	0.49	300
palestine	0.27	0.44	0.34	300
saudi	0.41	0.48	0.44	600
accuracy			0.72	16800
macro avg	0.60	0.64	0.61	16800
weighted avg	0.75	0.72	0.74	16800

```
samples = [
    "وين اقرب مطعم كشري؟",
    "وش الأخبار عندكم اليوم",
    "شنو أحسن قهوة في الدوحة؟"
]

samples_clean = [clean_text(s) for s in samples]


samples_tfidf = vectorizer.transform(samples_clean)


preds = clf.predict(samples_tfidf)


for s, p in zip(samples, preds):
    print("-" * 50)
    print(f"الجملة : {s}")
    print(f"التوقع  : {p}")
```

```
--------------------------------------------------
الجملة : وين اقرب مطعم كشري؟
التوقع  : Qatar
--------------------------------------------------
الجملة : وش الأخبار عندكم اليوم
التوقع  : saudi
--------------------------------------------------
الجملة : شنو أحسن قهوة في الدوحة؟
التوقع  : Qatar
```

7.2 MARBERT Results

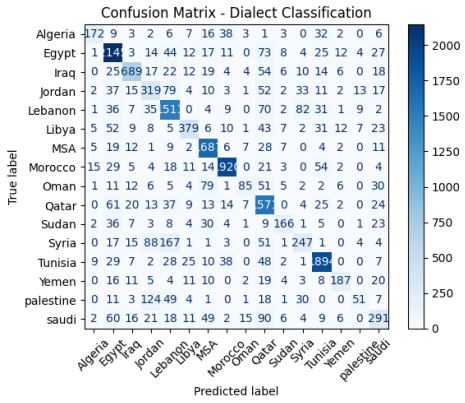
MARBERT performed significantly better:

- **Accuracy:** 79%
- **Macro F1:** 67%

The model demonstrated strong contextual understanding, enabling it to differentiate between dialects even in short or ambiguous sentences. MARBERT successfully captured subtle cues from morphology, spelling variations, and phrase structures.

The confusion matrix showed:

- Much fewer misclassifications
- Strong performance on Saudi, Egyptian, and most Gulf dialects
- Noticeably better handling of Maghrebi dialects compared to the ML model
- Overall improved generalization across all classes



The training and validation loss curves further confirm the strong performance of MARBERT. As shown in the figure, the training loss decreases steadily over the epochs, while the validation loss remains stable and close to it, indicating good

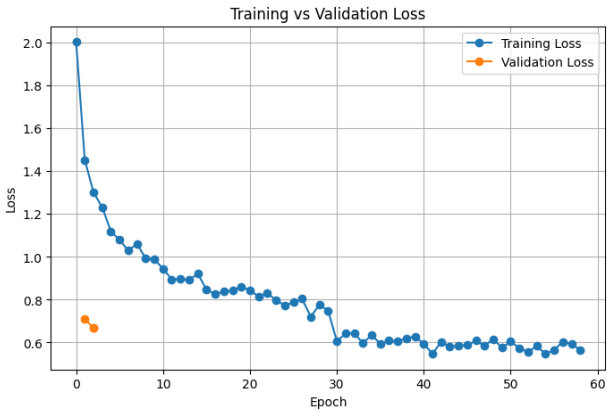
convergence and the absence of severe overfitting. At the end of training, the model achieved an **accuracy of 79%** and a **macro F1-score of 67%**, which supports the previously reported results.These results demonstrate that MARBERT learned rich contextual representations of Arabic dialects, enabling it to differentiate between dialects even in short or ambiguous sentences. The confusion matrix further supports this conclusion, showing fewer misclassifications, strong performance on Saudi, Egyptian, and Gulf dialects, and noticeably improved handling of Maghrebi dialects. Overall, the model exhibited strong generalization across all classes.

[9854/11900 19:51 < 04:07, 8.27 it/s, Epoch 1.66/2]

Epoch	Training Loss	Validation Loss	Accuracy	F1 Macro
1	0.776800	0.709213	0.776310	0.643422

[11900/11900 24:19, Epoch 2/2]

Epoch	Training Loss	Validation Loss	Accuracy	F1 Macro
1	0.776800	0.709213	0.776310	0.643422
2	0.565700	0.667120	0.792500	0.675088

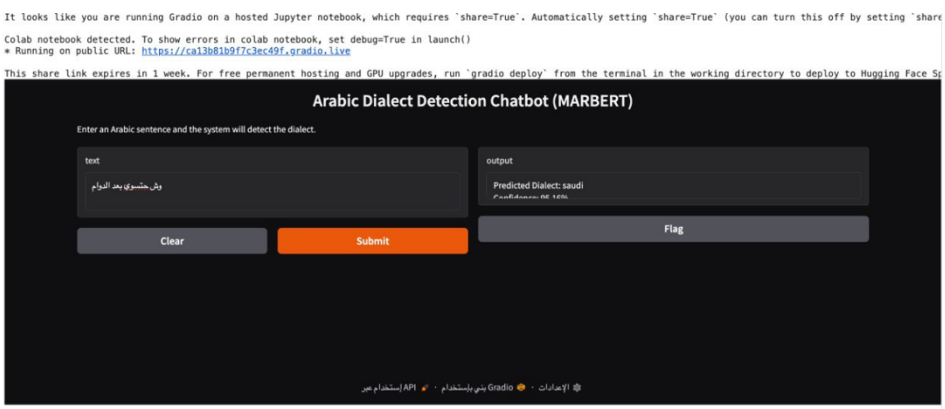


To qualitatively evaluate the trained model, a set of unseen Arabic sentences representing different dialects was tested. The model demonstrated a strong ability to correctly identify the corresponding dialects across multiple regions. It showed particularly high confidence in classifying Egyptian, Tunisian, Moroccan, Gulf, Iraqi, Sudanese, and Omani dialects. These results indicate that the model effectively captures dialectal patterns and linguistic variations, even in short and informal text inputs. This confirms the robustness and practical applicability of the proposed system.

```
Text: كيفاش
Predicted Dialect: Tunisia, Score: 0.4734
----------------------------------------
Text: شنية القيمة اليوم؟
Predicted Dialect: Tunisia, Score: 0.9960
----------------------------------------
Text: وين اطيق احول فلوسي؟
Predicted Dialect: Iraq, Score: 0.9911
----------------------------------------
Text: عافاك قول ليه ياماضا عيطات.
Predicted Dialect: Morocco, Score: 0.9972
----------------------------------------
Text: أبشر
Predicted Dialect: Qatar, Score: 0.3908
----------------------------------------
Text: هل تعيش بالقرب من اهني؟
Predicted Dialect: Oman, Score: 0.8874
----------------------------------------
Text: في زول بيساعدني. شكرا ليك.
Predicted Dialect: Sudan, Score: 0.9663
----------------------------------------
```

7.3 Interactive Web Interface (the MARBERT model)

A web-based interactive interface was developed using Gradio to allow real-time testing of the MARBERT model. The interface enables users to enter an Arabic sentence, after which the system automatically predicts the corresponding dialect and displays the prediction confidence.



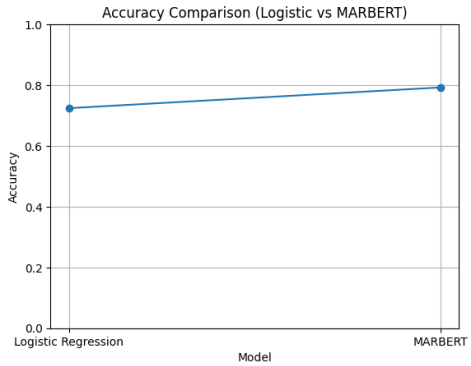
7.4 Comparative Interpretation

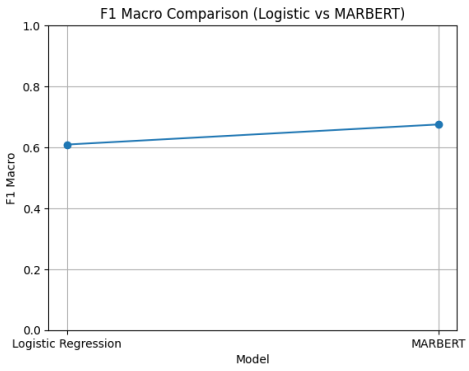
When comparing the two models side by side, MARBERT clearly outperformed the TF-IDF + Logistic Regression baseline in every metric. The gap between the two highlights the importance of contextual embeddings for Arabic dialect identification. While the TF-IDF + Logistic Regression model relies solely on surface-level word frequencies, MARBERT leverages deep semantic representation, enabling it to detect dialect markers even when they are subtle or distributed across the sentence. This confirms that transformer-based models are the current best approach for tasks involving dialectal Arabic.

	Model	Accuracy	F1 Macro	Training Type
0	Logistic Regression (TF-IDF)	0.7243	0.6091	Traditional Machine Learning
1	MARBERT	0.7925	0.6751	Deep Learning (Transformer)

The comparison results clearly show that MARBERT outperforms Logistic Regression in both Accuracy and Macro F1-score. This performance gap indicates that MARBERT is more capable of capturing the linguistic and Unlike TF-IDF with Logistic contextual differences between Arabic dialects. Regression, which relies mainly on surface word frequencies, MARBERT leverages deep contextual representations, allowing it to better handle short sentences, dialectal expressions, and spelling variations.

The higher Macro F1-score also reflects that MARBERT maintains more balanced performance across all dialect classes, even under class imbalance, whereas the Logistic Regression model struggled with underrepresented dialects. This confirms the effectiveness of transformer-based models for Arabic dialect classification tasks.





7.5 Summary of Findings

- Machine Learning provides a fast and interpretable baseline but lacks depth.
- MARBERT offers far superior accuracy and robustness across dialects.
- Deep contextual modeling is necessary for complex linguistic tasks such as dialect classification.
- Confusion matrices and evaluation plots clearly illustrate MARBERT's stronger performance and reduced error rates.

8. conclusion

In this project, we investigated the task of Arabic dialect identification using two distinct modeling approaches: a traditional Machine Learning method based on TF-IDF and Logistic Regression, and a Deep Learning method based on fine-tuning the MARBERT transformer model. The experimental results clearly demonstrate that deep contextual representations are essential for achieving high performance on dialect classification tasks.
The TF-IDF + Logistic Regression baseline provided reasonable results and served as an effective starting point. Its strengths lie in simplicity, fast computation, and the ability to capture frequent lexical markers. However, its reliance on surface-level word statistics limited its ability to accurately distinguish between dialects with shared vocabulary or similar syntactic patterns. The baseline model performed moderately well on dialects with strong lexical cues but struggled with short, ambiguous, or highly informal sentences
In contrast, the MARBERT model consistently outperformed the baseline across all evaluation metrics. Its ability to learn rich contextual and semantic features allowed it to handle morphological variation, orthographic diversity, and subtle dialect-specific patterns. MARBERT demonstrated stronger robustness across dialect families and significantly reduced misclassification between closely related dialects. These findings highlight the importance of transformer-based architectures for dialectal NLP tasks, particularly in high-variability linguistic environments such as Arabic.

9. References

[1] A. Alhaj Ali, "Text Classification for Arabic Dialects: Getting the Data," Kaggle, 2023. [Online]. Available: https://www.kaggle.com/code/ammarnassanalhajali/text-classification-arabic-dialect-getting-data/notebook

10. Appendix A: Google Colab Implementation

The full source code, data preprocessing steps, and interactive demo of the Arabic dialect detection system are available at the following Google Colab notebook: **https://colab.research.google.com/drive/17AFzrdWwsLayfIL_rEz0XM03upbiW8Lg?usp=sharing**